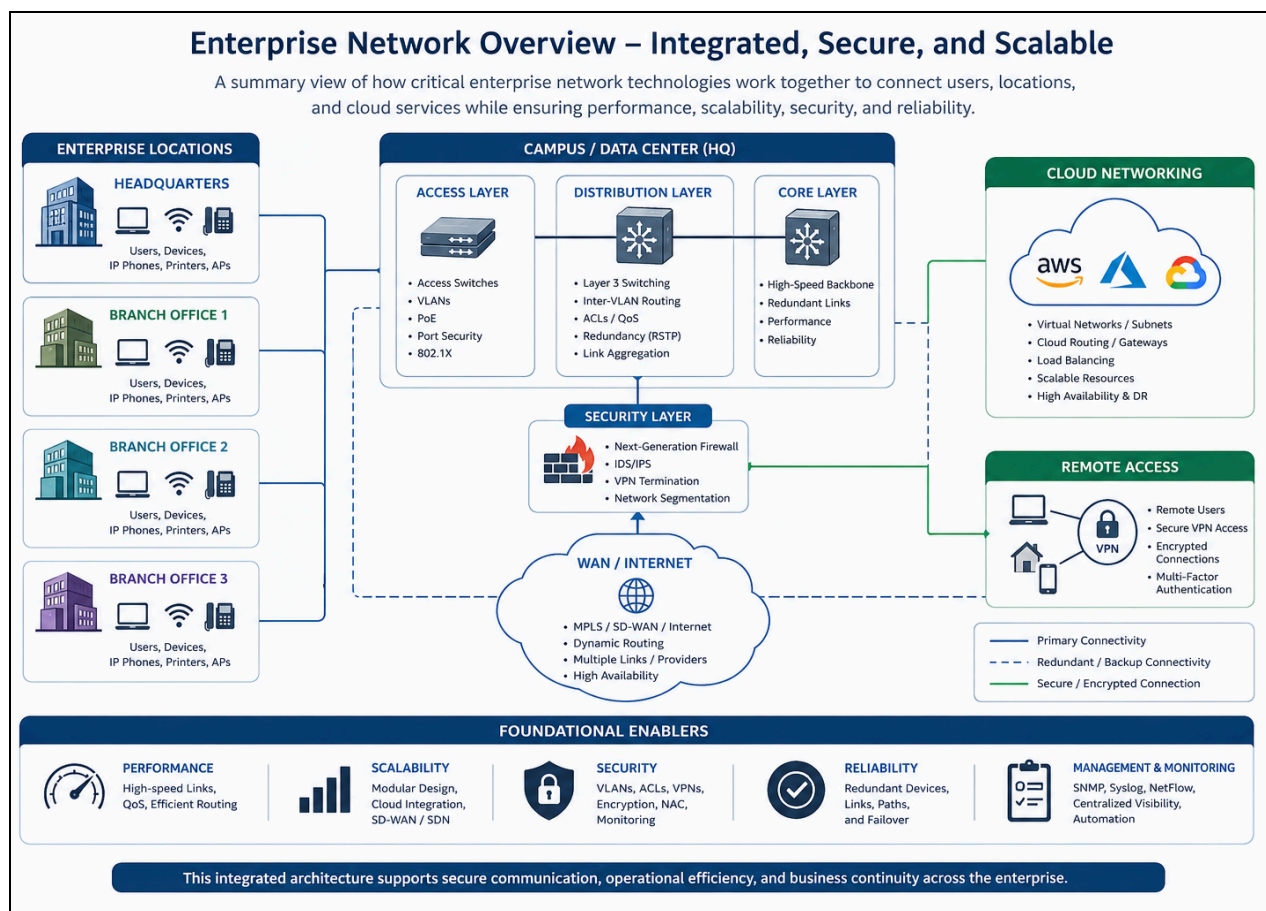


Milestone 6: Enterprise Network Technologies

Enterprise Network Technologies and Architecture

This enterprise network must provide secure, reliable, and scalable connectivity for headquarters, three branch offices, remote employees, and cloud-hosted business services. To meet these requirements, the architecture integrates LANs and WLANs for local connectivity; VLANs and enterprise switching for segmentation and traffic control; routing, WANs, and VPNs for communication between networks, locations, and remote users; and cloud networking for scalable access to business resources. The design also incorporates appropriate protocols and services, transmission media and transceivers, and resilient topologies and architectures. Together, these technologies form an integrated enterprise infrastructure that balances performance, scalability, security, and reliability while supporting efficient, continuous business operations.



Critical Enterprise Network Technologies

Enterprise networking combines network types and environments with the infrastructure and capabilities that make them function effectively. LANs, WLANs, VLANs, WANs, VPNs, and cloud networks establish connectivity, segmentation, and access across the enterprise, while technologies such as Ethernet switching, IP routing, security, QoS, redundancy, and monitoring provide the performance, control, protection, and resilience needed to operate those networks effectively. For this enterprise design, seven technologies are examined in greater detail because of their direct contribution to large-scale business operations.

Table 1. Critical Enterprise Network Technologies

Priority Technology	Role in the Enterprise	Why It Is Critical
1. LAN	Connects users, servers, printers, phones, access points, & other resources within each enterprise location.	Provides the high-speed local connectivity on which most daily enterprise communication depends.
2. Enterprise Switching	Provides Ethernet frame forwarding & supports VLANs, trunks, Layer 3 switching, redundancy, & access controls.	Forms the operational foundation of the enterprise LAN while supporting performance, segmentation, & resiliency.
3. VLAN	Logically separates departments, users, servers, voice, guests, & other functions within the switched network.	Improves segmentation, security, traffic management, & scalability without requiring separate physical networks.
4. WLAN	Extends LAN connectivity to laptops, mobile devices, scanners, & other wireless endpoints.	Provides secure mobility & flexible access throughout enterprise facilities.
5. WAN	Connects headquarters, branch offices, data centers, and other geographically separated networks.	Allows distributed locations to operate as one enterprise and access shared applications and services.
6. VPN	Creates encrypted site-to-site and remote-access connections across untrusted networks.	Protects data in transit and provides secure connectivity for remote employees and locations.
7. Cloud Networking	Provides virtual networks, subnets, routing, security, load balancing, and connectivity to cloud-hosted resources.	Provides elastic capacity, geographic availability, hybrid connectivity, and scalable application delivery.

Enterprise Technology Priorities: Performance, Scalability, Security, and Reliability

Each of the seven priority technologies contributes differently to the enterprise network. Together, they create an infrastructure that can perform efficiently, grow with the organization, protect business resources, and remain available when individual components or connections fail.

1. LAN — Local Area Network

The LAN provides the high-speed local foundation connecting employees, devices, and enterprise resources within each location.

- **Performance:** High-speed Ethernet provides low-latency connectivity for local users, applications, and services.
- **Scalability:** A hierarchical LAN design allows additional users, devices, switches, and network segments to be added as the organization grows.
- **Security:** NAC, port security, ACLs, and segmentation help control which devices connect and which resources they can access.
- **Reliability:** Redundant switches and links provide alternate paths when a device or connection fails.

2. Enterprise Switching

Enterprise switches connect LAN devices and provide the infrastructure required for VLANs, trunks, redundancy, access control, and Layer 3 connectivity.

- **Performance:** High-speed frame forwarding, QoS, and link aggregation support efficient delivery of business traffic.
- **Scalability:** Access, distribution, and core switching designs allow the network to expand in an organized hierarchy.
- **Security:** 802.1X, port security, VLANs, and ACLs provide multiple methods of controlling network access and traffic.
- **Reliability:** Rapid Spanning Tree Protocol (RSTP), Link Aggregation Control Protocol (LACP), redundant links, and redundant switches provide alternate paths and reduce single points of failure.

3. VLAN — Virtual Local Area Network

VLANs logically divide the switched LAN into separate broadcast domains, allowing departments and services to remain separated while sharing the same physical infrastructure.

- **Performance:** Smaller broadcast domains reduce unnecessary broadcast traffic and improve traffic organization.

- **Scalability:** New logical networks can be created without building separate physical networks for every department or function.
- **Security:** Finance, HR, IT, Voice, Servers, and Guest networks can be isolated, with inter-VLAN access controlled by ACLs or firewalls.
- **Reliability:** Segmentation can contain many Layer 2 problems within an individual VLAN rather than affecting the entire LAN. VLANs themselves do not provide redundancy; technologies such as RSTP, LACP, and redundant switches and links provide the actual failover mechanisms.

4. WLAN — Wireless Local Area Network

The WLAN extends enterprise LAN services to laptops, mobile devices, scanners, and other wireless endpoints while providing users with mobility throughout enterprise facilities.

- **Performance:** Proper AP placement, channel planning, coverage, and capacity provide efficient wireless connectivity for mobile users.
- **Scalability:** Additional access points can extend coverage and increase capacity as the number of users, devices, or facilities grows.
- **Security:** WPA2/WPA3-Enterprise, authentication, NAC, and VLAN segmentation help protect wireless access.
- **Reliability:** Overlapping AP coverage and redundant supporting infrastructure help maintain connectivity if an individual AP or path becomes unavailable.

5. WAN — Wide Area Network

The WAN connects headquarters, branch offices, data centers, and other geographically separated enterprise locations.

- **Performance:** Appropriate bandwidth, QoS, and routing allow business-critical applications, voice, and data to move efficiently between locations.
- **Scalability:** New branches, circuits, and network services can be incorporated as the enterprise expands geographically.
- **Security:** Firewalls, segmentation, routing controls, and VPNs protect traffic moving between enterprise locations and external networks.
- **Reliability:** Multiple service providers, redundant circuits, dynamic routing, and SD-WAN can provide alternate paths when a primary connection fails.

6. VPN — Virtual Private Network

VPNs provide encrypted connectivity across public or otherwise untrusted networks for remote employees and communications between enterprise locations.

- **Performance:** Properly sized VPN gateways securely transport remote-access and site-to-site traffic while maintaining acceptable application performance.

- **Scalability:** Additional users and locations can be securely connected without requiring dedicated private circuits for each connection.
- **Security:** Encryption protects data in transit, while authentication and MFA help ensure that only authorized users gain remote access.
- **Reliability:** Redundant VPN gateways and Internet connections can maintain secure connectivity when a primary gateway or connection becomes unavailable.

7. Cloud Networking

Cloud networking extends enterprise infrastructure into virtual environments where applications, networks, security controls, storage, and other services can be delivered without depending entirely on on-premises hardware.

- **Performance:** Distributed resources and load balancing can direct application traffic efficiently and reduce individual resource bottlenecks.
- **Scalability:** Virtual networks, compute resources, and services can expand or contract rapidly as business requirements change.
- **Security:** IAM, security groups, encryption, segmentation, and virtual firewalls provide layered controls for cloud resources.
- **Reliability:** Availability zones, replication, load balancing, backup, and disaster recovery strategies can keep services available when individual resources or locations fail.

Business Continuity and Efficiency

These technologies are most effective when they operate as parts of one architecture rather than as isolated solutions. For example, a Finance employee at a branch office could connect through the LAN or WLAN, enter the appropriate Finance VLAN, and have traffic forwarded through the enterprise switching infrastructure and Layer 3 routing toward headquarters or a cloud-hosted resource. When traffic crosses an untrusted network, a VPN can encrypt the connection before the employee reaches the application in the enterprise cloud environment.

Together, the seven technologies provide the enterprise with high-performing local and remote connectivity, scalable infrastructure, layered security, and redundant paths and services. These capabilities support business continuity by allowing employees to maintain access to critical applications, communications, and data as the organization grows and when individual network components or connections fail.

Configure Switching Technologies and Features

For this enterprise, the headquarters network should separate major business functions into VLANs instead of placing all devices into one broadcast domain.

Table 2. Headquarters VLAN Configuration

VLAN	Purpose	Example Subnet
10	Finance	10.10.10.0/24
20	Human Resources	10.10.20.0/24
30	Information Technology	10.10.30.0/24
40	Voice	10.10.40.0/24
50	Servers	10.10.50.0/24
60	Guest Wireless	10.10.60.0/24

Access ports would connect endpoint devices and place them into their assigned VLAN. 802.1Q trunk ports would carry multiple VLANs between switches. Because VLANs are separate Layer 2 broadcast domains, a Layer 3 switch or router would provide inter-VLAN routing. The switching configuration should also include:

- **RSTP** prevents Layer 2 loops while maintaining redundant paths and provides faster convergence after a topology change.
- **LACP link aggregation** between critical switches to combine links for additional bandwidth and redundancy.
- **802.1X authentication or port security** to control which devices can connect to enterprise switch ports.
- **PoE** for wireless access points, IP phones, and other supported devices.
- **ACLs** at Layer 3 boundaries to restrict communication between sensitive VLANs.
- **QoS** to prioritize latency-sensitive voice and video traffic.

For example, employees on the Guest VLAN should be allowed Internet access but blocked from Finance, HR, server, and management networks. Finance employees could access required enterprise servers while being restricted from unnecessary internal resources. This switching design provides segmentation, performance, controlled access, and redundancy while allowing the physical network infrastructure to support multiple departments efficiently.

Networking Ports, Protocols, Services, and Traffic Types

Enterprise administrators must understand the services traveling across the infrastructure so they can configure switches, routers, firewalls, monitoring systems, and troubleshooting tools correctly.

Table 3. Common Enterprise Ports, Protocols, and Services

Protocol / Service	Port(s)	Transport	Enterprise Function
SSH	22	TCP	Secure remote administration
DNS	53	TCP/UDP	Name resolution
DHCP	67/68	UDP	Automatic IP configuration
HTTP	80	TCP	Unencrypted web communication
NTP	123	UDP	Time synchronization
SNMP	161/162	UDP	Network monitoring and alerts
HTTPS	443	TCP	Secure web and application communication
SMB	445	TCP	File and printer sharing
RDP	3389	TCP/UDP	Remote Windows desktop access

TCP is connection-oriented and provides acknowledgments, sequencing, retransmission, and reliable delivery. It is appropriate for applications where complete delivery is important.

UDP is connectionless and has less overhead. It is useful for services where low latency or efficiency is more important than retransmitting every lost packet.

Administrators must also recognize common traffic delivery types:

Table 4. Common Enterprise Traffic Types

Traffic Type	Description	Example
Unicast	One sender to one destination	User accessing a web server
Broadcast	One sender to all devices in the local broadcast domain	ARP request
Multicast	One sender to a selected group of receivers	Streaming or routing updates
Anycast	Multiple systems provide the same service address, with routing directing traffic toward an appropriate instance	Distributed DNS or cloud services

Understanding these ports, protocols, services, and traffic types helps administrators troubleshoot connectivity and create firewall, ACL, QoS, and monitoring policies without unnecessarily blocking legitimate business traffic.

Compare Transmission Media and Transceivers

The physical medium must be selected according to required speed, distance, environment, cost, and reliability.

Table 5. Enterprise Transmission Media Comparison

Medium	Strengths	Limitations	Best Enterprise Use
Twisted-Pair Copper	Affordable, easy to install, supports Ethernet and PoE	Typically limited to 100 meters and can be affected by electromagnetic interference	Workstations, phones, printers, APs
Multimode Fiber (MMF)	High bandwidth and useful for shorter fiber connections	Shorter maximum distances than single-mode fiber	Building and data-center links
Single-Mode Fiber (SMF)	Very high bandwidth, long distance, low attenuation, resistant to electromagnetic interference	Optics and installation can cost more	Campus backbone, WAN, long-distance links
Wireless / RF	Mobility and flexibility without endpoint cabling	Interference, attenuation, shared bandwidth, and security considerations	Laptops, mobile devices, IoT

Copper is therefore appropriate for most horizontal endpoint connections, while fiber is better for enterprise backbones and high-speed switch-to-switch connections.

Transceivers

Modular transceivers allow switches and routers to support different media, speeds, and distances. Common families include:

- SFP — commonly supports 1-Gigabit Ethernet, depending on the module.
- SFP+ — commonly supports 10-Gigabit Ethernet.
- QSFP-family transceivers — support higher-capacity links by providing multiple high-speed channels.

The transceiver must be compatible with the switch interface, required data rate, fiber type, wavelength, connector, and transmission distance. For example, a short connection between switches inside a data center may use multimode fiber with compatible optics, while a long connection between enterprise buildings may require single-mode fiber and long-range optical transceivers.

The fastest transceiver is not automatically the correct choice; the administrator must match the entire physical link to the business requirement.

Network Topologies, Architectures, and Types

Enterprise networks combine several topology and architecture models rather than relying on one design everywhere.

Table 6. Network Topologies and Enterprise Applications

Topology	Advantages and Limitations	Enterprise Use
Star	Adv: Easy to manage and troubleshoot; endpoint failures are isolated Limit: Central switch can become a failure point without redundancy	Enterprise access LAN
Partial Mesh	Adv: Multiple paths improve resiliency Limit: More links, cost, and configuration complexity	Core, WAN, and critical infrastructure
Hub and Spoke	Adv: Efficient centralized branch connectivity Limit: Hub can become a bottleneck or failure point	Traditional WAN or VPN design
Point to Point	Adv: Simple and predictable direct connection Limit: Provides only one direct relationship per link	WAN and infrastructure links
Spine-Leaf	Adv: Multiple predictable high-speed paths and excellent east-west performance Limit: More appropriate for data centers than ordinary access networks	Modern data centers

For the proposed enterprise, a star topology is appropriate at the access layer because endpoint devices connect to centralized switches. Critical switches and WAN connections should have redundant or partially meshed paths so a single link failure does not disconnect the organization.

Enterprise Architecture

A traditional enterprise campus can use a hierarchical design where the access layer connects workstations, phones, printers, and wireless access points. The distribution layer aggregates access switches and commonly provides routing, policies, and redundancy. The core layer provides the high-speed backbone connecting major areas of the enterprise.

Smaller locations can use a collapsed-core architecture, combining the distribution and core functions. Data centers may instead use a spine-leaf architecture, where each leaf switch connects to each spine switch. This provides multiple predictable paths for high volumes of server-to-server, or east-west, traffic.

Table 7. Enterprise Network Types

Network Type	Scope / Purpose
LAN	Local wired network within a building or facility
WLAN	Wireless local network
CAN	Connects networks across a campus or group of nearby buildings
WAN	Connects geographically separated enterprise locations
SAN	Specialized network connecting servers and shared storage

A large enterprise commonly uses several simultaneously. Each branch may contain its own LAN and WLAN, headquarters may operate a campus network, the data center may contain a SAN, and the WAN connects all enterprise locations together.

The enterprise should primarily use a client-server architecture, with centralized or cloud-hosted services providing applications, authentication, storage, and business resources to users. A hybrid design can combine on-premises infrastructure with cloud networking when applications or workloads benefit from cloud scalability and geographic redundancy.

Enterprise Design Summary

The resulting enterprise architecture combines the technologies rather than treating them as separate solutions. Each technology provides distinct advantages and features at each layer of the OSI Model.

- At the **Physical Layer (Layer 1)**, copper, fiber, wireless media, and compatible transceivers provide connectivity.

- At the **Data Link Layer (Layer 2)**, VLANs, 802.1Q trunks, RSTP, LACP, PoE, port security, and 802.1X provide segmentation, redundancy, and access control.
- At the **Network Layer (Layer 3)**, routing, WAN technologies, VPNs, and cloud networking connect separate networks and locations.
- At the **Transport Layer (Layer 4)**, TCP and UDP provide end-to-end transport services for enterprise applications.
- At the **Application Layer (Layer 7)**, protocols and services such as DNS, DHCP, HTTP/HTTPS, SSH, NTP, SNMP, SMB, and RDP support enterprise applications and network operations.
- Across all layers, firewalls, ACLs, encryption, authentication, monitoring, redundancy, and failover protect business resources and maintain availability.

Conclusion

The most effective enterprise network is not simply the network with the greatest bandwidth. It must balance performance, scalability, security, and reliability.

LANs and WLANs provide local access, VLANs create logical segmentation, switching organizes local traffic, routing connects networks, WANs connect business locations, VPNs secure remote communications, and cloud networking expands the organization's infrastructure. Proper transmission media and transceivers support the physical network, while appropriate ports, protocols, topologies, and architectures allow the entire environment to operate efficiently.

Together, these technologies provide a scalable and resilient enterprise infrastructure capable of supporting employees, applications, communications, remote operations, and continued business operations even when individual links or network components fail.

<https://herbal-rancher.github.io/network-portfolio/>